Lecture 05 — Lookup & Reference Functions in Excel for Data Analytics

Unlike a basic single-table dataset, this workbook uses separate **transaction and master tables**, which reflects how business data is commonly structured.
The workbook contains:

Sheet	Business Purpose
Sales_Transactions	Main order-level transactional data
Product_Master	Product information, pricing, cost and inventory
Customer_Master	Customer details, segments and locations
Employee_Master	Employee and salesperson information
Monthly_Targets	Monthly sales targets in a horizontal structure

The primary concept to understand is simple:

A transaction table often stores IDs, while descriptive information is maintained separately in master tables. Lookup functions connect these tables.
For example, Sales_Transactions may contain only:
Product ID = PRD-0045
But the analyst may need:
Product Name
Category
Brand
Unit Price
Cost Price
Stock Quantity
These details can be retrieved from Product_Master using lookup functions.

Module 1: Understanding Cell Referencing in Excel

Before moving ahead to lookup functions, everyone needs to understand **cell references**, because lookup formulas depend heavily on selecting the correct lookup value, lookup range, return range, and table array.

1. What is a Cell Reference?

A **cell reference** is the address used to identify the location of a cell in an Excel worksheet.
Every cell address consists of:
Column Letter + Row Number
Examples:
A1
B5
D10
G100
In the Sales_Transactions sheet:

Cell	Possible Value
A2	ORD-100001
B2	Order Date
C2	Customer ID
D2	Product ID
E2	Salesperson ID

Therefore, if D2 contains:
PRD-0045
then a formula can use D2 as its lookup value.
Example:
=XLOOKUP(D2,Product_Master!A:A,Product_Master!B:B)
Here, D2 tells Excel:
Take the Product ID from cell D2 and search for it in the Product Master.

Module 2: Types of Cell References

Excel has three major types of cell references:

1. Relative Reference
2. Absolute Reference
3. Mixed Reference

Understanding these is essential for copying formulas correctly.

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2. Relative Cell Reference

A **relative cell reference changes automatically when a formula is copied or filled to another cell**.
Example:
=A2*B2
If copied one row down, Excel automatically changes it to:
=A3*B3
Then:
=A4*B4
The references adjust relative to the position of the formula.

Business Example: Calculating Gross Sales

Suppose we retrieve the Unit Price from the Product Master and add it to column K.
The transaction table could look like this:

Product ID	Quantity	Unit Price	Gross Sales
PRD-0045	3	₹8,999	?
PRD-0012	5	₹18,999	?

Formula:
=F2*K2
Here:

- F2 = Quantity
- K2 = Unit Price

When copied downward:
=F3*K3
=F4*K4
=F5*K5
This is exactly where **relative references** should be used because every transaction needs to use values from its own row.

Business Use Case

Relative references are useful for:

- Revenue calculations
- Profit calculations
- Discount calculations
- Tax calculations
- Commission calculations
- Row-by-row classifications

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3. Absolute Cell Reference

An **absolute reference remains fixed when a formula is copied to another cell**.
It uses the $ symbol before both the column letter and row number.
Syntax:
$A$1
This means:

- Column A is locked.
- Row 1 is locked.

Even when copied anywhere, $A$1 remains $A$1.
Shortcut
Press:
F4
to cycle through the different reference types.

Business Case: Applying a Fixed GST Rate

Suppose:
N1 = GST Rate
N2 = 18%
And M2 contains the sales amount.
Formula:
=M2*$N$2
When copied downward:
=M3*$N$2
=M4*$N$2
=M5*$N$2
Notice:

- M2 changes because every row has a different sales amount.
- $N$2 remains fixed because every transaction uses the same GST rate.

Why is absolute referencing required?

If you write:
=M2*N2
and copy it downward, Excel changes the formula to:
=M3*N3
=M4*N4
This is wrong because the GST rate exists only in N2.
The correct formula is:
=M2*$N$2

Business Case: Sales Commission

Suppose:
P2 = Commission Rate = 5%
And M2 contains Net Sales.
Formula:
=M2*$P$2
The commission rate stays fixed while the sales value changes for each transaction.

Business Case: Currency Conversion

Suppose:
R2 = USD to INR Exchange Rate = 86.50
And M2 contains sales in USD.
Formula:
=M2*$R$2
This converts every sales transaction using one fixed exchange rate.

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4. Mixed Cell Reference

A **mixed reference locks either the row or the column, but not both**.
There are two types:
$A1
Column locked, row changes.
And:
A$1
Row locked, column changes.
Comparison

Reference	Column	Row
A1	Changes	Changes
$A$1	Locked	Locked
$A1	Locked	Changes

Reference	Column	Row
A$1	Changes	Locked

Business Case: Monthly Sales Target Matrix

The Monthly_Targets sheet contains months horizontally:

Employee ID	Jan	Feb	Mar	Apr
EMP-001	500000	520000	550000	580000
EMP-002	450000	480000	510000	540000

Suppose you are creating calculations across multiple employees and months. Mixed references can ensure that:

- The Employee ID column remains fixed when copying horizontally.
- The Month header row remains fixed when copying vertically.

For example:
=$A2
locks column A while allowing the row to change.
And:
=B$1
locks row 1 while allowing the column to change.
This becomes particularly useful in:

- Two-dimensional lookup matrices
- Financial models
- Sales forecasting
- Budget allocation
- Price-volume matrices
- Sensitivity analysis

Module 3: Introduction to Lookup Functions

What is a Lookup?

A **lookup operation searches for a specific value in one location and returns related information from another location**.
For example:
Input:
Product ID = PRD-0001
Required Output:
Product Name = Wireless Mouse Pro
The lookup function performs the connection:
PRD-0001 → Search Product Master → Return Wireless Mouse Pro

Why are Lookup Functions Important in Data Analytics?

Consider the Sales_Transactions table:

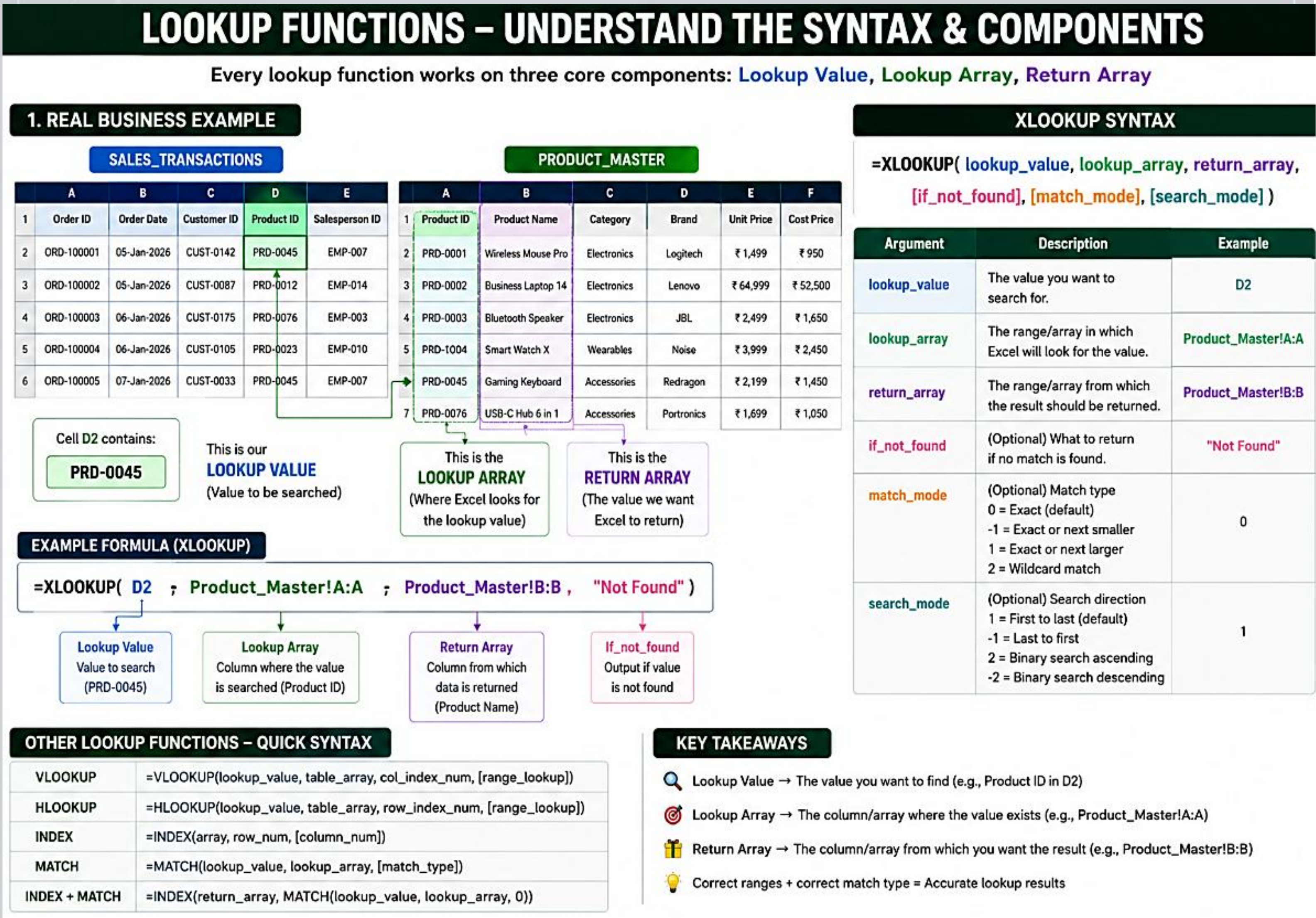
Order ID Customer ID Product ID Salesperson ID

ORD-100001 CUST-0142 PRD-0045 EMP-007

An analyst cannot present a report containing only IDs. The business needs meaningful information such as:

Customer Name
Product Name
Category
Brand
Salesperson Name
Region
Manager

Lookup functions allow analysts to enrich transaction data using master tables.



Module 4: VLOOKUP Function

VLOOKUP searches for a value in the **first column of a table** and returns a corresponding value from another column in the same row.

VLOOKUP stands for:

Vertical Lookup

It searches vertically from top to bottom.

Syntax
=VLOOKUP(lookup_value,table_array,col_index_num,[range_lookup])

Syntax Explanation

Argument	Meaning
lookup_value	Value you want to search for
table_array	Complete table containing the lookup and return columns
col_index_num	Position number of the return column within the selected table
[range_lookup]	FALSE for exact match or TRUE for approximate match

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Business Example 1: Find Product Name Using Product ID

Suppose D2 in Sales_Transactions contains:
PRD-0045
The Product_Master table contains:

A	B	C	D	E	F
Product ID	Product Name	Category	Sub-Category	Brand	Unit Price

Formula:
=VLOOKUP(D2,Product_Master!$A$2:$J$81,2,FALSE)
Formula Breakdown
D2
The Product ID we want to find.
Product_Master!$A$2:$J$81
The complete lookup table. It is made absolute because the master table should remain fixed when the formula is copied downward.
2
Return data from the second column of the selected table: Product Name.
FALSE
Require an exact match.

Business Example 2: Retrieve Product Category

=VLOOKUP(D2,Product_Master!$A$2:$J$81,3,FALSE)
The column index is 3 because Category is the third column.

Business Example 3: Retrieve Unit Price

=VLOOKUP(D2,Product_Master!$A$2:$J$81,6,FALSE)
The sixth column contains Unit Price.

Business Example 4: Retrieve Customer Name

The Customer ID is in C2.
Formula:
=VLOOKUP(C2,Customer_Master!$A$2:$J$421,2,FALSE)

This retrieves the customer's name.

Business Example 5: Retrieve Customer Region

=VLOOKUP(C2,Customer_Master!$A$2:$J$421,6,FALSE)
This returns the corresponding customer's region.

Why Use Absolute Referencing in VLOOKUP?

Consider:
=VLOOKUP(D2,Product_Master!A2:J81,2,FALSE)
When copied one row downward, Excel may change the table range to:
=VLOOKUP(D3,Product_Master!A3:J82,2,FALSE)
This is a problem because the master table is shifting.
Therefore, use:
=VLOOKUP(D2,Product_Master!$A$2:$J$81,2,FALSE)
Now:

- D2 changes to D3, D4, D5, etc.
- $A$2:$J$81 remains permanently fixed.

This is a perfect example of combining **relative and absolute references in the same formula**.

Exact Match vs Approximate Match

Exact Match — FALSE

Use:
FALSE
when you need an exact ID, code, email, product name, or employee identifier.
Example:
=VLOOKUP(D2,Product_Master!$A$2:$J$81,2,FALSE)
Use exact matching for:

- Product IDs
- Customer IDs
- Employee IDs
- Invoice numbers
- Order IDs
- Account numbers

Approximate Match — TRUE

Approximate matching is used for ranges or bands.
For example:

Minimum Sales	Rating
0	Poor
100000	Average
300000	Good

Minimum Sales	Rating
500000	Excellent

Formula:
=VLOOKUP(A2,$F$2:$G$5,2,TRUE)
If:
A2 = 350000
Result:
Good

Critical Rule

For approximate VLOOKUP, the first column should generally be sorted in ascending order.
Do not use approximate matching carelessly with IDs.

Limitations of VLOOKUP

This is an important interview topic.

1. Cannot naturally look to the left

The lookup column must be the first column in the selected table.

2. Uses hard-coded column numbers

Example:
=VLOOKUP(D2,A:J,6,FALSE)
The 6 is manually specified.
If columns are inserted or rearranged, formulas can return incorrect results or break.

3. Exact match is not the default

If the fourth argument is omitted, VLOOKUP defaults to approximate matching, which can produce unexpected results.

4. Returns only the first matching record

If duplicates exist, VLOOKUP returns the first matching value.

Module 5: HLOOKUP Function

HLOOKUP searches horizontally across the **first row of a table** and returns a corresponding value from another row.
HLOOKUP stands for:
Horizontal Lookup

Syntax

=HLOOKUP(lookup_value,table_array,row_index_num,[range_lookup])

Syntax Explanation

Argument	Meaning
lookup_value	Value to find in the first row
table_array	Horizontal data range
row_index_num	Row number from which to return a result
[range_lookup]	FALSE for exact match or TRUE for approximate match

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Business Example: Monthly Sales Targets

Suppose the Monthly_Targets sheet is structured like this:

Metric	Jan	Feb	Mar	Apr
EMP-001	500000	520000	550000	580000
EMP-002	450000	480000	510000	540000

If B1 contains:
Mar
Formula:
=HLOOKUP(B1,Monthly_Targets!$B$1:$M$26,2,FALSE)
This searches the first row for Mar and returns the corresponding value from row 2.

Important Limitation

HLOOKUP is not the best choice when **both employee and month must be dynamic**. For that scenario, a two-way INDEX + MATCH + MATCH lookup is more flexible.
That distinction is worth teaching because students should understand **when a function is appropriate**, not just how its syntax works.

Module 6: XLOOKUP Function

XLOOKUP searches for a value in one range and returns a corresponding result from another range.
It is the modern replacement for many traditional VLOOKUP and HLOOKUP scenarios.

Syntax
=XLOOKUP(lookup_value,lookup_array,return_array,[if_not_found],[match_mode],[search_mode])
Syntax Explanation

Argument	Meaning
lookup_value	Value to search
lookup_array	Range containing the value to search
return_array	Range containing the result
[if_not_found]	Custom output when no match exists
[match_mode]	Controls exact, approximate, or wildcard matching
[search_mode]	Controls search direction or algorithm

Business Example 1: Find Product Name

=XLOOKUP(D2,Product_Master!$A$2:$A$81,Product_Master!$B$2:$B$81,"Product Not Found")
Explanation
D2
Product ID to find.
Product_Master!$A$2:$A$81
Where Excel should search.
Product_Master!$B$2:$B$81
What Excel should return.

"Product Not Found"
Custom output if the Product ID doesn't exist.

Business Example 2: Retrieve Unit Price

=XLOOKUP(D2,Product_Master!$A$2:$A$81,Product_Master!$F$2:$F$81,"Product Not Found")

Business Example 3: Find Employee Manager

Suppose E2 contains the Salesperson ID.
=XLOOKUP(E2,Employee_Master!$A$2:$A$26,Employee_Master!$F$2:$F$26,"Employee Not Found")
This retrieves the manager responsible for that salesperson.

Business Example 4: Left Lookup

Suppose you know the Product Name but want the Product ID.
VLOOKUP struggles because Product ID is to the left of Product Name.
With XLOOKUP:
=XLOOKUP(A2,Product_Master!$B$2:$B$81,Product_Master!$A$2:$A$81,"Not Found")
XLOOKUP can search and return in either direction.

Business Example 5: Multiple Column Return

Modern versions of Excel can return multiple columns:
=XLOOKUP(D2,Product_Master!$A$2:$A$81,Product_Master!$B$2:$F$81,"Not Found")
This can return:

- Product Name
- Category
- Sub-Category
- Brand
- Unit Price

with one formula using dynamic arrays.

XLOOKUP vs VLOOKUP

Feature	VLOOKUP	XLOOKUP
Exact match by default	No	Yes
Left lookup	No	Yes
Right lookup	Yes	Yes
Custom not-found message	Requires error handling	Built in
Hard-coded column index	Yes	No
Multiple-column return	Limited	Yes
Search from last to first	No	Yes
Modern flexibility	Lower	Higher

Interview Answer

XLOOKUP is generally more flexible than VLOOKUP because it separates the lookup array from the return array, supports left and right lookups, uses exact match by default, provides built-in error handling, and doesn't rely on hard-coded column index numbers.

Module 7: MATCH Function

The MATCH function searches for a value within a range and returns its **relative position**, not the actual value itself.

Syntax

=MATCH(lookup_value,lookup_array,[match_type])

Match Types

Value	Meaning
0	Exact match
1	Exact or next smaller value
-1	Exact or next larger value

For IDs and exact business records, normally use:

0

Business Example

Suppose:

D2 = PRD-0045

Formula:

=MATCH(D2,Product_Master!$A$2:$A$81,0)

If PRD-0045 is the 45th item in the selected range, Excel returns:

45

MATCH does not return the Product Name. It returns the position.

This becomes powerful when combined with INDEX.

Module 8: INDEX Function

The INDEX function returns a value from a specified row and optionally column within a range.

Syntax

=INDEX(array,row_num,[column_num])

Syntax Explanation

Argument	Meaning
array	Range containing the data
row_num	Row position to return
[column_num]	Column position, if required

Simple Example

Suppose:

=INDEX(Product_Master!$B$2:$B$81,5)

This returns the fifth value from the Product Name range.

But manually writing 5 is not dynamic.

That's why INDEX is commonly combined with MATCH.

Module 9: INDEX + MATCH

INDEX + MATCH combines two functions:

- MATCH finds the position of the required record.
- INDEX returns the actual value from that position.

Conceptually:

MATCH → Where is the record?

INDEX → What value exists at that position?

Business Example: Find Product Name

=INDEX(Product_Master!$B$2:$B$81,MATCH(D2,Product_Master!$A$2:$A$81,0))

Step 1: MATCH

=MATCH(D2,Product_Master!$A$2:$A$81,0)

Suppose this returns:

45

Step 2: INDEX

Excel effectively performs:

=INDEX(Product_Master!$B$2:$B$81,45)

And returns the corresponding Product Name.

Business Example: Retrieve Cost Price

=INDEX(Product_Master!$G$2:$G$81,MATCH(D2,Product_Master!$A$2:$A$81,0))

This searches for the Product ID and returns Cost Price.

Business Example: Find Customer Loyalty Tier

=INDEX(Customer_Master!$I$2:$I$421,MATCH(C2,Customer_Master!$A$2:$A$421,0))

This retrieves the customer's loyalty tier.

Business Example: Find Employee Manager

=INDEX(Employee_Master!$F$2:$F$26,MATCH(E2,Employee_Master!$A$2:$A$26,0))

This returns the employee's manager.

Module 10: Two-Way Lookup Using INDEX + MATCH + MATCH

This is where lookup logic becomes more advanced and business-oriented.
Suppose the Monthly_Targets sheet contains:

Employee ID	Jan	Feb	Mar	Apr
EMP-001	500000	520000	550000	580000
EMP-002	450000	480000	510000	540000

You want to dynamically answer:
What was the sales target for EMP-002 in March?
This requires two searches:

1. Find the employee row.
2. Find the month column.

Formula:
=INDEX(Monthly_Targets!$B$2:$M$26,
MATCH(A2,Monthly_Targets!$A$2:$A$26,0),
MATCH(B2,Monthly_Targets!$B$1:$M$1,0))
Where:
A2 = EMP-002
B2 = Mar
Logic
First MATCH:
=MATCH(A2,Monthly_Targets!$A$2:$A$26,0)
Finds the employee's row position.
Second MATCH:
=MATCH(B2,Monthly_Targets!$B$1:$M$1,0)
Finds the month's column position.
INDEX returns the value at the intersection.
This is called a:
Two-Way Lookup

Module 11: Combining Lookups with Previously Learned Functions

A good Data Analytics course shouldn't teach functions in isolation. Students should combine the concepts from previous lectures.

Example 1: Calculate Employee Tenure After Lookup

=DATEDIF(XLOOKUP(E2,Employee_Master!$A$2:$A$26,Employee_Master!$E$2:$E$26),TODAY(),"Y")
This:

1. Finds the employee's joining date.
2. Calculates completed years of tenure.

Example 2: Calculate Gross Sales Using Retrieved Unit Price

=F2*XLOOKUP(D2,Product_Master!$A$2:$A$81,Product_Master!$F$2:$F$81,0)

This:

1. Takes Quantity from F2.
2. Finds Unit Price using Product ID.
3. Calculates Gross Sales.

Example 3: Calculate Net Sales After Discount

=(F2*XLOOKUP(D2,Product_Master!$A$2:$A$81,Product_Master!$F$2:$F$81,0))*(1-G2)

Where:

- F2 = Quantity
- D2 = Product ID
- G2 = Discount %

This creates an actual business calculation rather than a lookup exercise with no analytical purpose.

Example 4: Classify High-Value Orders

=IF(F2*XLOOKUP(D2,Product_Master!$A$2:$A$81,Product_Master!$F$2:$F$81,0)>100000,"High Value","Regular")

This combines:

- IF
- XLOOKUP
- Multiplication
- Business classification

Common Lookup Errors

#N/A

Meaning:

No matching record was found.

Possible reasons:

- Incorrect Product ID
- Missing master record
- Extra spaces
- Text-number mismatch
- Typing error

Using XLOOKUP:

=XLOOKUP(D2,Product_Master!$A$2:$A$81,Product_Master!$B$2:$B$81,"Product Not Found")

#REF!

Usually means the formula references an invalid or deleted cell or column.

For example, a VLOOKUP formula can break if its return-column structure is altered incorrectly.

#VALUE!

Can occur because of:

- Invalid argument types
- Incompatible range dimensions
- Incorrect formula construction

Important Interview Questions

1. What is the difference between relative and absolute references?
A relative reference changes when copied, while an absolute reference remains fixed using $.

2. Why do we use $ in lookup formulas?
To prevent the lookup table or reference range from shifting when the formula is copied.

3. What is the difference between VLOOKUP and XLOOKUP?
VLOOKUP requires the lookup value to be in the first column of its table array and relies on a numerical column index. XLOOKUP separates lookup and return arrays, supports lookups in either direction, uses exact matching by default, and includes built-in not-found handling.

4. What is the difference between INDEX and MATCH?
MATCH finds a position. INDEX returns a value from a position.

5. Why combine INDEX and MATCH?
MATCH dynamically finds the position, and INDEX returns the corresponding value.

6. Can VLOOKUP perform a left lookup directly?
No, not in its standard form. XLOOKUP and INDEX+MATCH can.

7. What is a two-way lookup?
A lookup where both the row and column are determined dynamically.

8. What happens if duplicate lookup values exist?
Standard VLOOKUP, XLOOKUP with default search mode, and INDEX+MATCH typically return the first matching occurrence.

9. Why is FALSE important in VLOOKUP?
It forces an exact match.

10. When would you use HLOOKUP?
When lookup values are arranged horizontally in the first row of a table.